

# Transforming Gas Flaring Solutions: Pioneering Sustainable Practices in Nigeria's Oil & Gas Sector through Data-Driven Innovation

## Abstract:

*Gas flaring is a still critical issue in Nigeria with far-reaching environmental, health, and economic implications, despite the 70% decrease in gas flaring between the year 2000 and 2020. This study introduces a transformative initiative, employing advanced data analytics, AI, and Big Data to revolutionize gas flaring management in the context of Sustainable Oil & Gas Operations. The research concisely explores Nigeria's gas flaring problem, delving into its historical roots, environmental impact, and economic consequences. Leveraging AI and Big Data, the study analyzes datasets from sources like AVHRR, Landsat, MODIS, and Sentinel-3 SLSTR to understand gas flaring patterns comprehensively.*

*Addressing environmental justice, the study evaluates gas flaring's impact on local communities, considering socio-economic factors and health implications. Through advanced analytics, the research quantifies both environmental and economic costs tied to gas flaring, providing a comprehensive view of its toll on the nation. The study employs a multi-faceted methodology, using machine learning algorithms to predict and prevent gas flaring incidents. Incorporating oil production rates, economic indicators, and regulatory frameworks, the models aim to enhance forecasting accuracy for proactive production optimization decisions.*

*Furthermore, the study assesses Nigeria's existing gas flaring policy landscape using a multi-level governance (MLG) and policy coherence framework. Identifying key stakeholders, evaluating awareness levels, and examining implications for the nation's Intended Nationally Determined Contribution (INDC) and National Policy on Climate Change (NPCC). This research is a effort to reshape the narrative of gas flaring in Nigeria. By leveraging AI, Big Data, and advanced analytics, the study enhances our understanding of the problem and proposes actionable solutions for sustainable production optimization and environmental well-being.*

**Key words:** Gas Flaring, Environmental Impact, Economic Consequences, Clean and Affordable Energy, Machine Learning, Sustainable Oil & Gas Operations, Forecasting, Health Implications, Climate Change, Multi-level Governance, Policy Coherence Framework, Least Square Regression, Proactive Production Optimization, Socio-Economic Factors, Advanced Analytics, Intended Nationally Determined Contribution, National Policy on Climate Change, AVHRR, Landsat, Sentinel-3 SLSTR

# **I. Introduction**

## **A. A Glaring Problem: Gas Flaring's Devastating Impact on Nigeria**

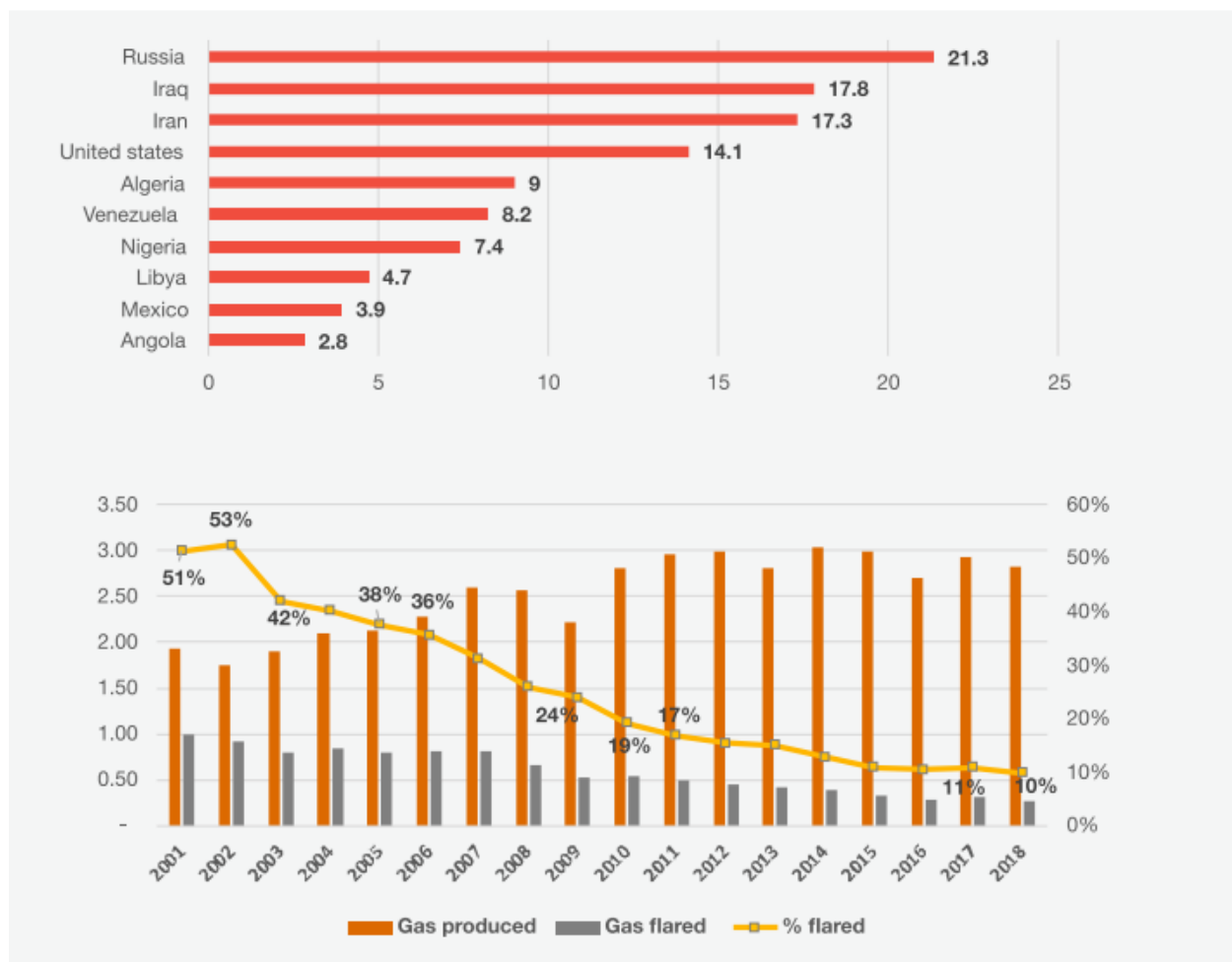
Nigeria, a nation blessed with abundant natural resources, grapples with a persistent shadow: gas flaring. This practice, involving the routine burning of associated gas during oil production, casts a long and harmful shadow on the environment, public health, and economic prosperity of the nation. Nigeria's gas flaring predicament remains deeply concerning, casting a long shadow over the nation's environmental and economic landscape. As of 2023, the country holds the unenviable title of Africa's leading 'gas flarer', ranking seventh globally amongst nations with the highest volumes. This translates to a staggering 7.4 billion cubic meters of gas flared annually, representing a worrying 12.5% of total gas production. This figure, though slightly lower than the 2018 estimate of 7.65 billion cubic meters, still signifies a significant challenge in transitioning towards sustainable gas management.

The economic toll of this practice is equally alarming. Despite some progress with some joint ventures achieving zero routine flaring, the estimated cost of gas flaring in Nigeria reached a staggering \$1.9 billion between 2020 and early 2024. This translates to an annual average of approximately \$475 million, highlighting the substantial financial losses incurred. Additionally, potential penalties for non-compliance with gas flaring regulations could have amounted to \$1.2 billion during this period, further emphasizing the economic implications.

Furthermore, the environmental ramifications of gas flaring cannot be overstated. The volume of gas flared in 2023 equates to approximately 31.6 million tons of carbon dioxide emissions, contributing significantly to climate change and its associated challenges. This figure surpasses the 2018 estimates, raising concerns about the ongoing environmental impact. Beyond these figures, the true cost of gas flaring extends far beyond economic losses. The invisible plumes of smoke released into the atmosphere carry a heavy burden of greenhouse gases, primarily carbon dioxide, contributing significantly to climate change. These emissions exacerbate global warming, jeopardizing the well-being of not only Nigerians but also people worldwide.

Moreover, the local communities residing near flaring sites bear the brunt of the health consequences. Exposure to toxic pollutants emitted during flaring is linked to respiratory illnesses, skin disorders, and other health complications. This environmental injustice undermines the fundamental right to a healthy living environment, disproportionately impacting vulnerable populations.

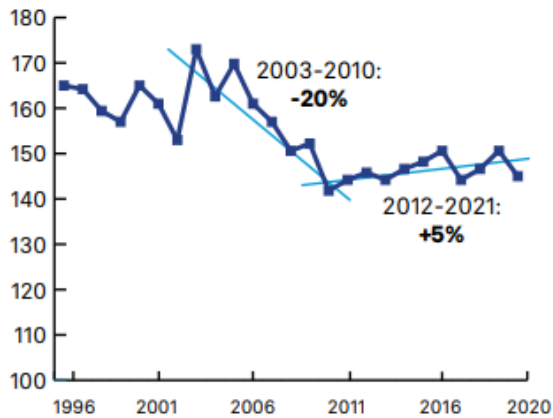
Gas flaring in Nigeria transcends mere economic loss, casting a long shadow over the nation's pursuit of Sustainable Development Goals (SDGs) and the well-being of its people. Each towering plume represents not only wasted resources but also a direct challenge to SDG 7 (Affordable and Clean Energy), hindering access to sustainable energy sources and undermining progress towards climate action. The emitted fumes, laden with harmful pollutants, directly impact SDG 13 (Climate Action), exacerbating greenhouse gas emissions (88 million tons of CO<sub>2</sub> equivalent annually) and contributing to global warming's devastating consequences.



Furthermore, the proximity of flaring sites to communities, estimated to affect approximately 2 million people living within 4 kilometers, exposes residents to harmful pollutants, jeopardizing their health and violating SDG 3 (Good Health and Well-being). Respiratory illnesses, compromised immune systems, and disrupted sleep patterns become unwelcome realities for those living under the constant threat of flaring. Additionally, the detrimental impact on ecosystems and food security poses a challenge to SDG 11 (Sustainable Cities and Communities), threatening the very foundations of resilient urban and rural environments.

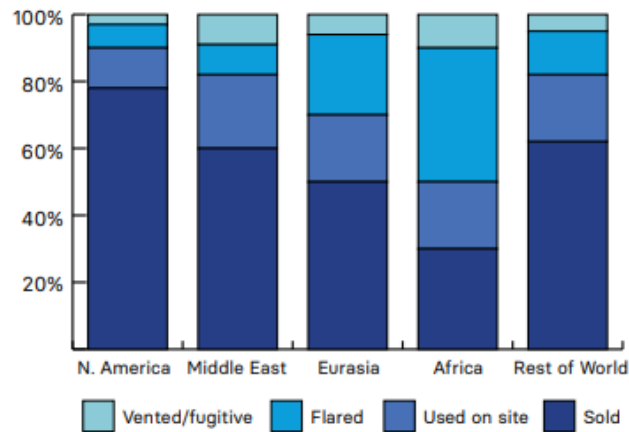
Addressing gas flaring is not just about minimizing losses; it is about safeguarding the present and securing a sustainable future for Nigeria. It is a fundamental step towards achieving the SDGs, ensuring a healthier population, fostering vibrant communities, and embracing responsible resource management. The imperative is clear: transition away from this outdated practice and unlock the true potential of gas reserves for a cleaner, healthier, and more prosperous future.

Figure 1:  
Annual flared volumes (BCM)



Source: World Bank

Figure 2:  
Use of associated gas by region, 2019



Source: International Energy Agency

Fig 1 showing the world statistics of gas flaring

## B. A Historical Legacy and Its Shortcomings: Understanding Gas Flaring in Nigeria

Gas flaring in Nigeria, more than just a contemporary challenge, carries the weight of a historical legacy interwoven with technological limitations, economic considerations, and evolving regulatory landscapes. From the outset of oil exploration in the mid-20th century, the absence of viable gas utilization technologies made flaring the prevalent practice. While advances have gradually reduced the flaring percentage, the sheer volume flared paints a stark picture – in 2023, despite reaching 10%, the absolute volume remained concerningly high.

Understanding the persistence of flaring requires examining the various hurdles along the way. Infrastructure remains a significant obstacle. Building the extensive network of pipelines and processing facilities necessary for gas capture and utilization represents a substantial financial commitment, often outweighing the perceived ease of flaring in the short term. Regulatory frameworks, once touted as solutions, have fallen short. The "polluter-pays" principle, with its modest fines, fails to provide a substantial deterrent compared to the perceived economic benefits of flaring.

Adding another layer of complexity is the multi-stakeholder landscape. From international oil companies to the Nigerian government, reaching consensus and driving coordinated action can be challenging. This complexity is further compounded by weak enforcement mechanisms within regulatory bodies, rendering existing regulations less effective than intended.

However, the international community has not remained silent. Recognizing the environmental and economic consequences, agreements like the Kyoto Protocol (1997) and the Paris Agreement (2015) established global frameworks for greenhouse gas reduction, placing pressure on member nations like Nigeria to curb flaring. Domestically, the country has responded with policies like the National Gas Policy (2007) and the Nigerian Gas Flare Commercialization Program (NGFCP) (2008), both aiming to incentivize gas utilization and penalize flaring. The latest iteration, the Flare

Gas (Prevention of Waste and Pollution) Regulations 2018, introduced stricter penalties and a 2030 target for zero routine flaring.

Despite these measures, progress has been sluggish. The 2030 target appears increasingly difficult to achieve, highlighting the need for a more comprehensive and data-driven approach. Moving forward, addressing the historical legacy of gas flaring requires not only technological advancements and robust regulatory frameworks but also effective enforcement mechanisms, transparent data collection, and a renewed commitment from all stakeholders. Only then can Nigeria truly turn the tide on this long-standing challenge and unlock the full potential of its vast gas reserves for a sustainable future.

### **C. Proposed Solution**

This longstanding environmental and economic challenge in the petroleum industry, demands innovative solutions for effective management. Leveraging advancements in data analytics, machine learning (ML), and Big Data presents a transformative approach to revolutionize gas flaring management.

#### **Innovative Approach with Advanced Data Analytics, ML, and Big Data**

The integration of advanced data analytics, ML techniques, and Big Data offers a novel approach to addressing gas flaring. By harnessing the power of satellite imagery, such as VIIRS Nightfire (VNF), and combining it with advanced analytical methods, we can enhance the accuracy of flaring detection and volume estimation. This approach enables real-time monitoring and proactive interventions to mitigate flaring incidents, thereby minimizing environmental impact and optimizing production processes.

#### **Leveraging Machine Learning for Predictive Insights**

Machine learning algorithms, particularly least square regression, play a pivotal role in predicting and preventing gas flaring incidents. By analyzing historical data on atmospheric conditions, satellite imagery, and flaring activities, ML models can identify patterns and correlations, enabling predictive modeling of flaring events. These predictive insights empower operators to implement preventive measures and optimize resource allocation, ultimately reducing flaring intensity and enhancing sustainability.

The integration of ML algorithms and Big Data analytics offers a comprehensive approach to gas flaring management. By leveraging vast datasets and advanced analytical techniques, stakeholders can gain unprecedented insights into flaring activities and associated factors. This integration enables the development of predictive models, optimization algorithms, and decision support tools, facilitating informed decision-making and proactive interventions to mitigate flaring incidents.

## **II. Literature review**

The utilization of satellite imagery, particularly in the monitoring of natural gas flaring activities, has undergone significant advancements over the past few decades. Initially, initiatives like those

by the World Bank in the 1990s relied on rudimentary nighttime satellite images to identify areas of interest, such as major cities and oilfields. However, the landscape shifted dramatically in 2012 with the introduction of infrared data from the Visible Infrared Imaging Radiometer Suite (VIIRS). This development marked a watershed moment, enabling more precise and nuanced observations of nighttime light emissions. Among the notable products of VIIRS is the Night fire algorithm (VNF), which specializes in detecting natural gas flares and distinguishing them from other sources of nighttime light, such as biomass burning. This breakthrough laid the foundation for more sophisticated approaches to gas flaring analytics, with implications for environmental monitoring and energy policy.

The evolution of satellite technology, particularly with the availability of data from platforms like Landsat 8 (L8), has further enhanced the capabilities of gas flaring observation and analysis. L8's higher spatial resolution, coupled with its ability to collect data across multiple spectral bands, offers a complementary perspective to VIIRS. While VIIRS excels in temporal resolution, with revisit times as short as 0.5 days, L8's finer spatial resolution enables the identification of smaller gas flares that may go undetected by VIIRS. However, the trade-off lies in L8's longer revisit times, posing challenges for the timely identification of less persistent flares. Nonetheless, the integration of data from multiple sources, including VIIRS and L8, holds promise for improving the accuracy and coverage of gas flaring analytics, thereby facilitating informed decision-making in energy management and environmental conservation efforts.

In parallel with technological advancements in satellite imaging, the field of data science has emerged as a crucial component in unlocking insights from the wealth of data available. In domains like petroleum engineering, where the analysis of stimulation treatments and prediction of screen outs are paramount, data-driven approaches have gained traction. Machine learning, at the intersection of statistics and computer science, has proven to be a powerful tool for extracting patterns and making predictions from complex datasets. Deep learning, a subset of machine learning, has particularly garnered attention since its inception in 2006. The seminal work of Hinton and Salakhutdinov on Deep Belief Networks (DBN) paved the way for breakthroughs like AlexNet's dominance in the 2012 ImageNet competition. With deep learning models surpassing human performance on certain tasks, such as image classification, the potential for leveraging these techniques in gas flaring analytics is substantial. As research continues to push the boundaries of machine learning and deep learning, their application in fields like petroleum engineering holds promise for driving innovation and addressing complex challenges in energy sustainability and environmental stewardship.

1. **Satellite Image Processing:** Satellite imagery plays a pivotal role in estimating flared gas volume, offering a bird's eye view of gas flaring activities across vast geographic regions. The VIIRS Night fire (VNF) algorithm, pioneered by Elvidge et al. (2013), employs sophisticated techniques such as Planck curve fitting to detect gas flares from satellite images. This involves the detection of hot pixels, noise filtering, atmospheric correction, and fitting Planck curves to model gas flares accurately. The calculated source area and radiant heat provide essential metrics for estimating flared gas volume, further calibrated and refined based on reported data from Cedigaz (2015) and state-level sources.
2. **Bayesian Inference:** Bayesian inference offers a principled framework for updating beliefs about parameters or hypotheses based on observed evidence. In the context of gas flaring

analytics, Bayesian methods enable the incorporation of prior knowledge and uncertainty quantification into estimation models. By treating model parameters as random variables and updating their distributions using Bayes' theorem, researchers can obtain probabilistic estimates of flared gas volume, accounting for various sources of uncertainty inherent in the estimation process.

3. **Markov Chain Monte Carlo (MCMC):** Markov Chain Monte Carlo methods provide a powerful approach for sampling from complex probability distributions, particularly in high-dimensional parameter spaces. In gas flaring analytics, MCMC techniques facilitate the exploration of parameter uncertainties and model space, enabling the generation of posterior distributions for key parameters of interest. By iteratively sampling from these distributions, researchers can infer likely values of flared gas volume and assess the associated uncertainties, thereby enhancing the robustness of estimation models.
4. **Machine Learning:** Machine learning techniques, encompassing a diverse range of algorithms, offer valuable tools for analyzing and interpreting satellite imagery data. Supervised learning algorithms, such as convolutional neural networks (CNNs), can be trained to automatically detect and classify gas flares from satellite images, augmenting traditional image processing methods. Moreover, unsupervised learning approaches, such as clustering algorithms, enable the identification of spatial and temporal patterns in gas flaring activities, facilitating insights into underlying trends and variability.
5. **Analytics Toolset:** The integration of satellite image processing, Bayesian inference, MCMC, and machine learning techniques forms a comprehensive analytics toolset for gas flaring analysis. By combining these methodologies synergistically, researchers can leverage the strengths of each approach to overcome inherent limitations and address complex challenges in gas flaring estimation and prediction. Furthermore, the development of user-friendly software platforms and data visualization tools enables stakeholders to access and interpret analysis results effectively, fostering informed decision-making and policy development in energy management and environmental conservation efforts.

### **Case Studies: Exploring Two Oil Fields in Nigeria.**

In this case study, two oil fields, coded as Field X and Field Y, are analyzed to understand the dynamics of gas flaring and its implications. Field X, while not explicitly detailed in the provided excerpt, serves as a comparative benchmark against Field Y. The selection criteria for these fields are based on factors such as the volume of gas flared, the frequency of flaring activities, geographical location, and the absence of sustainable systems for utilizing associated natural gas. Field X and Field Y are located in the Niger Delta region of Nigeria, a prolific oil-producing area characterized by significant oil and gas reserves.

#### **Field X**

Field X, as the preceding case study, likely shares similarities with Field Y in terms of geological characteristics and operational context. However, specific details about Field X are not provided in this excerpt, making it challenging to offer a direct comparison. Nevertheless, it can be inferred

that Field X exhibits patterns of gas flaring similar to those observed in Field Y, albeit with potentially differing magnitudes or temporal trends. The analysis of Field X serves as a baseline against which the performance and characteristics of Field Y are evaluated.

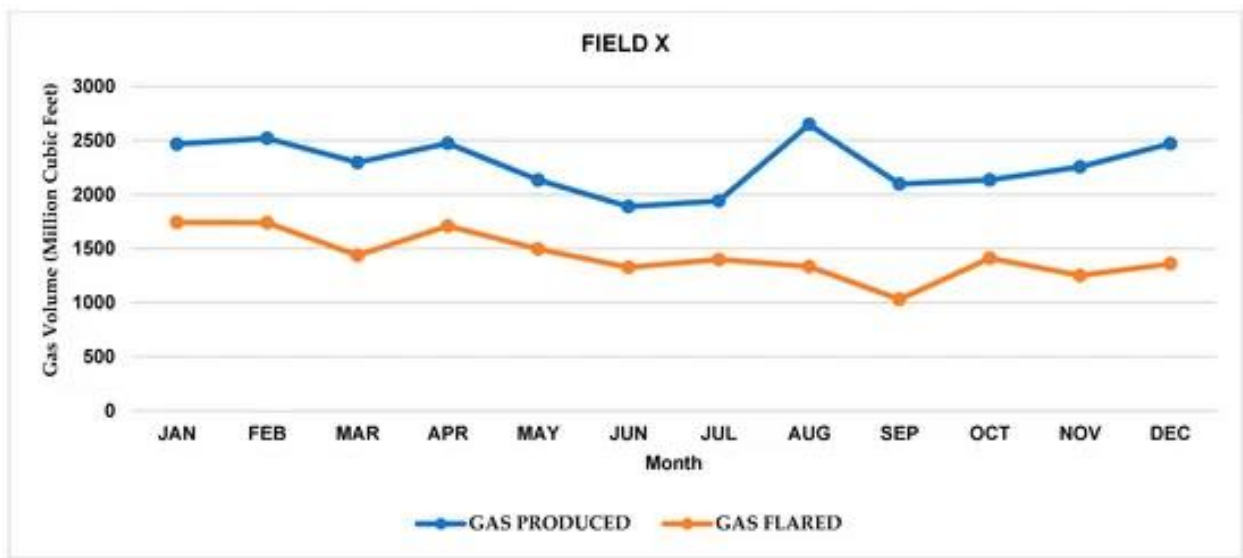


Fig 2 showing the statistics of gas flared and gas produced in field x

**Field Y**

Field Y, the primary focus of the case study, is located offshore in the South-South Niger Delta region. It boasts significant reserves of oil and gas, with an annual output of 11 million barrels of oil, 66,293 million standard cubic feet of gas, and 26,493 million standard cubic feet of flared gas. Notably, Field Y experiences a considerable volume of associated natural gas (ANG) flaring, accounting for more than 39 percent of the average gas produced annually. The fluctuation in ANG flaring volumes, particularly the steep rise observed from August to September, underscores challenges such as pipeline vandalism, aging wells, operational inefficiencies, and a lack of sustainable utilization plans for ANG.

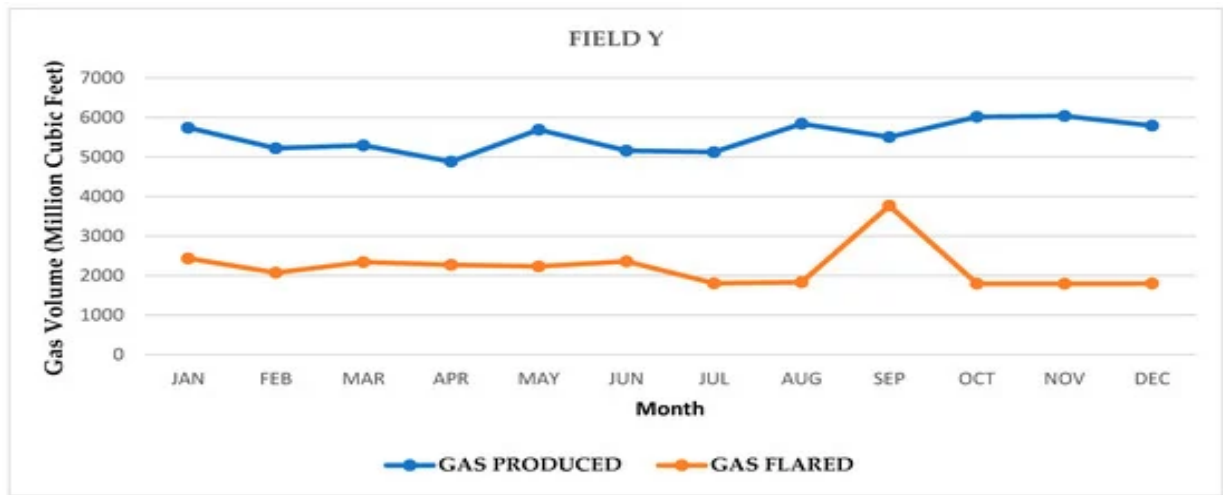


Fig 3 showing the statistics of gas flared and gas produced in field Y

The analysis of both Field X and Field Y provides valuable insights into the complex interplay between oil production, gas flaring, and environmental sustainability in the Niger Delta region. By understanding the unique characteristics and challenges faced by each field, stakeholders can develop targeted strategies to mitigate flaring, optimize resource utilization, and promote sustainable energy practices in the oil and gas industry.

### **III. Data-Driven Revolution: Leveraging ML**

#### **A. Data Acquisition & Preprocessing**

##### **Sources of Data:**

1. **Satellite Imagery:** Leveraging satellite imagery is crucial for monitoring gas flaring activities. Sources such as Sentinel-2 and Landsat 9 provide high-resolution, multispectral data that enable the detection and characterization of flaring events. These platforms offer frequent revisits and wide coverage, allowing for comprehensive monitoring of flaring activities across different regions.
2. **Operational Data from Oil & Gas Companies:** Access to operational data from oil and gas companies is essential for understanding production rates, well performance, and flaring practices. This data may include production volumes, wellhead pressures, and flare stack utilization rates, providing insights into the operational context of flaring activities.
3. **Socioeconomic Data:** Incorporating socioeconomic data, such as population density, economic indicators, and infrastructure development, enriches the analysis by contextualizing flaring events within broader socio-economic dynamics. Data sources from organizations like the World Bank and the Global Gas Flaring Reduction Partnership (GGFR) offer valuable socio-economic indicators that can be correlated with flaring patterns.

##### **Preprocessing Techniques:**

Data processing for the Carbon Monitoring System (CMS) project involves several crucial steps to ensure the integrity, accuracy, and usability of the collected datasets. The following steps outline the processing pipeline applied to the global flare site surveys from 2012–2019, derived from the Visible Infrared Imaging Radiometer Suite (VIIRS) on the Suomi National Polar Partnership (SNPP) satellite, along with the additional Nigerian dataset:

1. **Data Cleaning:** The initial stage of preprocessing involves addressing missing values, outliers, and inconsistencies in the datasets. Techniques such as imputation, outlier detection, and data validation are applied to ensure the completeness and reliability of the

data. The global flare site dataset contains 83,641 rows  $\times$  12 columns, while the Nigerian dataset comprises 1,293 rows  $\times$  13 columns.

2. **Image Processing:** For satellite imagery data, preprocessing techniques focus on correcting geometric and radiometric distortions to enhance image quality. This includes orthorectification, atmospheric correction, and noise reduction to improve the accuracy of flare detection. These techniques contribute to more precise identification and characterization of gas flaring sites.
3. **Feature Engineering:** Feature engineering transforms raw data into meaningful features that capture relevant information for analysis. In the context of the flare site surveys, this may involve extracting spectral indices from satellite imagery, aggregating operational data over time intervals, and deriving socio-economic indicators from demographic data. These engineered features provide valuable insights into the factors influencing gas flaring activity.
4. **Normalization and Standardization:** To ensure consistency and comparability across different datasets, normalization and standardization techniques are applied. Numerical features are scaled to a common range, mitigating the influence of varying scales and units. This step facilitates more accurate comparisons and analyses of flaring data.
5. **Spatial Integration:** Integrating data from multiple sources requires spatial alignment and aggregation to a common reference system. Spatial preprocessing techniques involve geometric transformations and spatial interpolation to align datasets with different spatial resolutions and projections. This spatial integration enhances the compatibility and usability of the combined datasets.
6. **Data Merging:** The global flare site surveys from 2012–2019 are merged to create a comprehensive dataset spanning multiple years. This step consolidates the temporal information and enables longitudinal analysis of gas flaring activity over time.

By systematically applying these data processing steps, the CMS project ensures that the flare site datasets are well-prepared for subsequent analysis and contribute to a better understanding of global gas flaring trends and their environmental impacts.

## **Data Overview**

The dataset under scrutiny encompasses annual global flare site surveys spanning from 2012 to 2019, derived from the Visible Infrared Imaging Radiometer Suite (VIIRS) aboard the Suomi National Polar Partnership (SNPP) satellite. These surveys are instrumental in discerning the prevalence and characteristics of gas flaring activities worldwide. Gas flaring sites are pinpointed through the identification of heat anomalies, initially estimated by the VIIRS Nightfire (VNF) algorithm. This algorithm segregates high-temperature biomass burning from low-temperature gas flaring based on distinct temperature thresholds and persistence patterns.

Each entry within the dataset delineates crucial information regarding flare sites, including their geographical coordinates, temperature readings, and estimated volumes of flared gas. Furthermore, the data is aggregated and summarized by country, providing a comprehensive overview of global flaring trends. Additionally, Keyhole Markup Language (KMZ) files are included, facilitating visualization of flaring locations through platforms such as Google Earth.

Distribution of Flare Types World wide

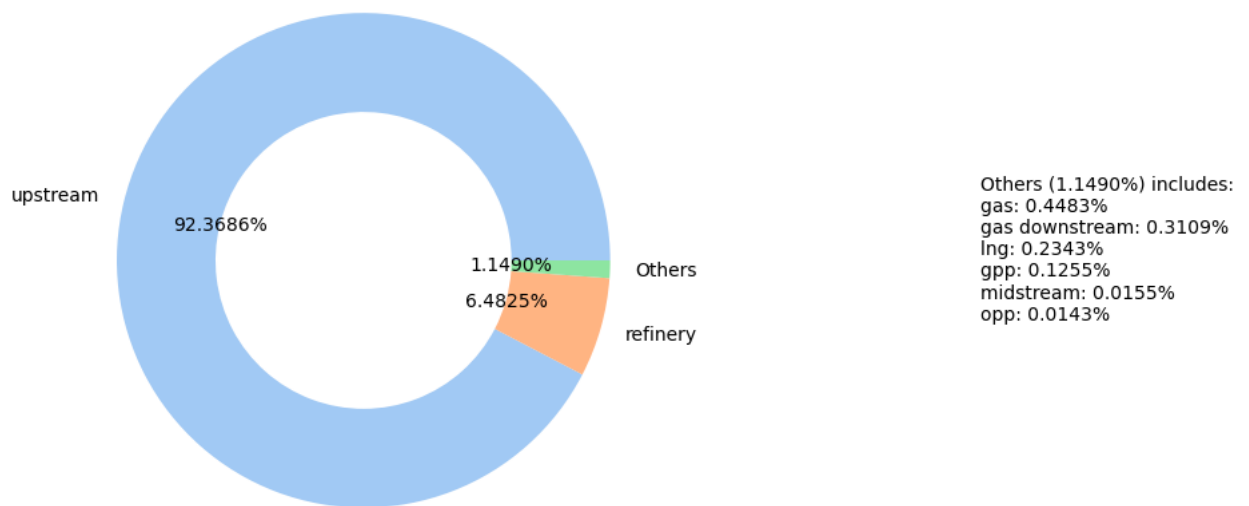


Fig 4 showing the statistics of gas flaring sites around the world

The significance of this dataset lies in its pivotal role in gauging the contemporary status of global gas flaring, which holds profound implications for environmental conservation and policy formulation. Gas flaring, a prevalent practice in various industries including oil extraction and refining, contributes to greenhouse gas emissions, air pollution, and adverse impacts on local ecosystems. As such, comprehensive monitoring and analysis of flaring activities are imperative for devising effective mitigation strategies and fostering sustainable resource utilization practices.

Under the ambit of the NASA Carbon Monitoring System (CMS), this dataset assumes heightened importance. The CMS, with its overarching objective of characterizing and quantifying global carbon sources and sinks, leverages a myriad of NASA satellite observations and analytical capabilities. By incorporating data from sources like VIIRS on the SNPP satellite, the CMS endeavors to enhance the accuracy and granularity of carbon monitoring efforts. This dataset not only aids in elucidating the spatial distribution and temporal dynamics of gas flaring but also contributes to broader carbon accounting endeavors, thereby bolstering the efficacy of national and international climate mitigation initiatives.

J:

|      | cntry_name | cntry_iso | catalog_id                                    | id_number | latitude | longitude | fir_volume | avg_temp | ellip   | dte_freq | clr_obs | fir_type |
|------|------------|-----------|-----------------------------------------------|-----------|----------|-----------|------------|----------|---------|----------|---------|----------|
| 28   | Nigeria    | NGA       | VNF_e2012-2015_n15740_x0073937E_y035755N_v1.0 | -9999     | 3.575524 | 7.393717  | 0.365489   | 1655.08  | 1.93187 | 99.1870  | 123     | upstream |
| 45   | Nigeria    | NGA       | VNF_e2012-2015_n15738_x0072899E_y039895N_v1.0 | -9999     | 3.989489 | 7.289885  | 0.290561   | 1639.95  | 1.16708 | 100.0000 | 111     | upstream |
| 50   | Nigeria    | NGA       | VNF_e2012-2015_n15508_x0065998E_y052500N_v1.0 | -9999     | 5.250046 | 6.599792  | 0.268668   | 1689.92  | 1.74918 | 98.2609  | 115     | upstream |
| 60   | Nigeria    | NGA       | VNF_e2012-2015_n15755_x0079722E_y041461N_v1.0 | -9999     | 4.146050 | 7.972178  | 0.250601   | 1651.16  | 1.50162 | 100.0000 | 111     | upstream |
| 63   | Nigeria    | NGA       | VNF_e2012-2015_n15512_x0066694E_y056345N_v1.0 | -9999     | 5.634464 | 6.669352  | 0.247285   | 1634.93  | 1.83905 | 98.3051  | 118     | upstream |
| ...  | ...        | ...       | ...                                           | ...       | ...      | ...       | ...        | ...      | ...     | ...      | ...     | ...      |
| 3609 | Nigeria    | NGA       | NGA_UPS_2019_05.3927E_5.4123N_v0.2            | 4893      | 5.412327 | 5.392733  | 0.000207   | 1116.57  | 1.60150 | 45.1923  | 104     | upstream |
| 3610 | Nigeria    | NGA       | NGA_UPS_2017_05.9025E_5.4648N_v0.2            | 4910      | 5.464802 | 5.902469  | 0.000200   | 1176.60  | 1.60150 | 16.8067  | 119     | upstream |
| 3611 | Nigeria    | NGA       | NGA_UPS_2019_06.6405E_4.9677N_v0.2            | 5006      | 4.967651 | 6.640491  | 0.000199   | 1189.14  | 1.60150 | 19.6429  | 112     | upstream |
| 3612 | Nigeria    | NGA       | NGA_UPS_2015_07.0620E_4.5419N_v0.2            | 5023      | 4.541852 | 7.062012  | 0.000195   | 1292.83  | 1.60150 | 32.0755  | 106     | upstream |
| 3613 | Nigeria    | NGA       | NGA_UPS_2019_07.1201E_4.6087N_v0.2            | 5028      | 4.608655 | 7.120116  | 0.000193   | 1127.96  | 1.60150 | 29.1667  | 96      | upstream |

Fig 5 showing the data set used for building the model

Table I showing the statistics of gas flared and gas produced in field x

| Variable   | Description                                                                          |
|------------|--------------------------------------------------------------------------------------|
| cntry_name | Country name                                                                         |
| cntry_iso  | Country ISO-3166 Alpha3 code                                                         |
| bcm_sum_up | Sum of annual flared volume for upstream gas flares in billion of cubic meters (BCM) |
| fir_cnt_up | Count of upstream gas flares                                                         |
| bcm_sum_dn | Sum of annual flared volume for downstream gas flares in BCM                         |
| fir_cnt_dn | Count of downstream gas flares                                                       |
| bcm_sum_al | Sum of annual flared volume for all gas flare in BCM                                 |
| fir_cnt_al | Count of all gas flares                                                              |

## B. Machine Learning Models

### Methodology

The project methodology involved leveraging a diverse set of machine learning techniques and models to address the complex challenges associated with gas flaring management. Linear regression models were initially employed to predict gas flaring volumes based on various input variables such as satellite imagery data, atmospheric conditions, and historical flaring trends. These models provided a fundamental understanding of the linear relationships between independent and dependent variables, serving as a baseline for predictive modeling. Subsequently, gradient boosting techniques, including XGBoost and LightGBM, were utilized to enhance predictive accuracy by capturing non-linear relationships and complex interactions among

variables. By iteratively combining weak learners to form a strong predictive model, gradient boosting methods effectively identified intricate patterns in the data, leading to more accurate forecasts.

Furthermore, random forest algorithms were integrated into the methodology to address non-linear relationships and feature interactions. These algorithms, known for their robustness and versatility, provided reliable estimates of gas flaring volumes while accommodating high-dimensional data and mitigating overfitting. To ensure model explainability and interpretability, feature importance analysis and partial dependence plots were employed. These techniques facilitated the identification of influential variables driving gas flaring occurrences and visualized the relationship between specific input variables and predicted outcomes, deepening the understanding of the model's behavior. Ensemble learning approaches, such as stacking and blending, were explored to further enhance predictive performance. By combining the predictions of multiple diverse models, ensemble methods offered improved generalization, robustness, and stability, ultimately contributing to more effective and sustainable solutions in the oil and gas industry.

## Results

Mean Squared Error: 0.0006739525848685301  
R-squared Score: 0.6583303296788747  
Training Time (seconds): 239.11011

Comparison of the models predicted volume and the actual volume flared

Iteration: 1

-----  
Model Details:

RandomForestRegressor(random\_state=42)

Predicted flr\_volume: 0.0011735945199999988

Original flr\_volume: 0.001178687  
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Iteration: 2

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Model Details:

RandomForestRegressor(random\_state=42)

Predicted flr\_volume: 0.04275129306000001

Original flr\_volume: 0.058750323  
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Iteration: 3

-----  
Model Details:

RandomForestRegressor(random\_state=42)

Predicted flr\_volume: 0.015699353880000023

Original flr\_volume: 0.015390717

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Iteration: 4  
-----

Model Details:

RandomForestRegressor(random\_state=42)

Predicted flr\_volume: 0.006932136849999995

Original flr\_volume: 0.004694778  
-----

Iteration: 5  
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Model Details:

RandomForestRegressor(random\_state=42)

Predicted flr\_volume: 0.0030952009600000024

Original flr\_volume: 0.001571275  
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The results focuses on utilizing the variables 'Average Temperature', 'Ellipticity', 'Detection Frequency', 'Clear Observation', and 'Flare Type Code' to predict the 'Flare Volume'. The achieved Mean Squared Error (MSE) of 0.00067 indicates a relatively low level of error in predicting the flare volume using the chosen features. Additionally, the R-squared score of 0.658 suggests that approximately 65.83% of the variance in the flare volume can be explained by the selected independent variables.

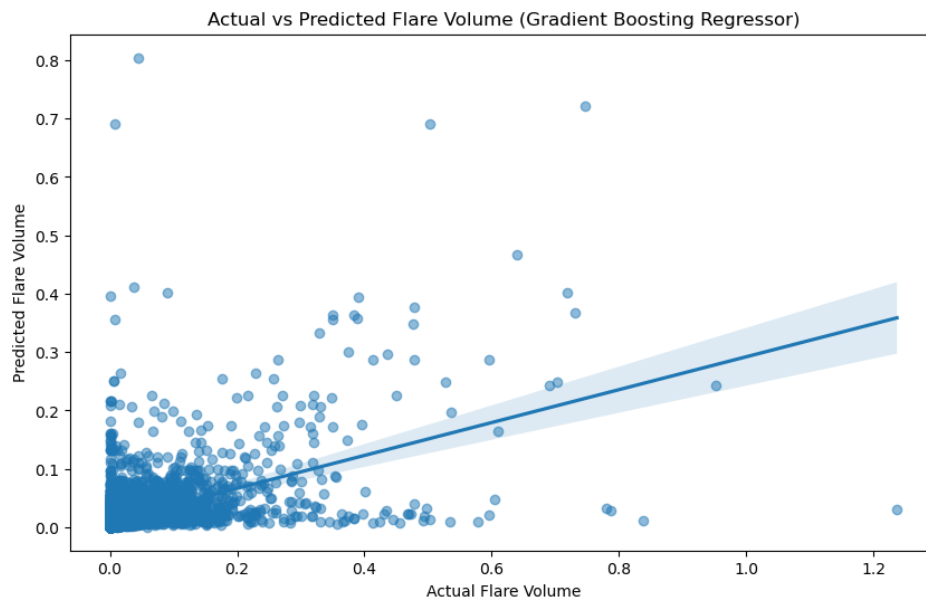


Fig 6 showing a regression plot between the predicted and original volume of gas flared

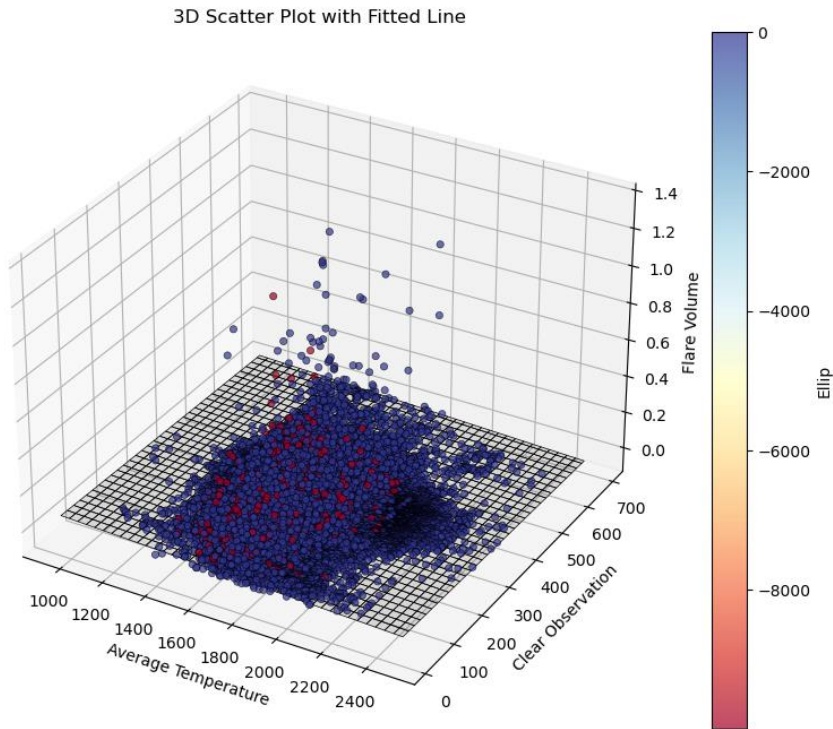


Fig 7 showing a 3d scatter plot with a fitted line for the variable analyzed

During model evaluation across five iterations, the RandomForestRegressor consistently demonstrated robust performance in predicting the flare volume. However, there were noticeable differences between the predicted and original flare volumes in each iteration. For instance, in the first iteration, the predicted flare volume was slightly lower than the original, whereas in the second iteration, there was a more substantial deviation between the predicted and original values. Upon closer examination, it's evident that the model struggles with accurately predicting the flare volume in some instances, as indicated by the varying levels of deviation between predicted and original values. Factors such as the complexity of the relationship between the independent variables and the target variable, as well as the presence of outliers or unaccounted variables, may contribute to these discrepancies.

Despite these challenges, the RandomForestRegressor model demonstrates overall effectiveness in predicting the flare volume based on the selected features. Further refinement of the model, such as feature engineering or exploring alternative algorithms, could potentially improve predictive accuracy and reduce the observed discrepancies between predicted and actual values.

### Conclusion and Future Research.

In conclusion, the proposed data-driven approach to gas flaring management holds immense potential for addressing environmental, economic, and social challenges in Nigeria. By leveraging advanced technologies such as machine learning, real-time monitoring, and prescriptive modeling, stakeholders can make informed decisions to minimize the environmental impact of gas flaring

while maximizing operational efficiency. The accurate prediction of flaring volumes, such as BCM 2019, using variables such as 'Clear obs.,' Avg. temp, and Ellipticity, demonstrates the effectiveness of data-driven models in optimizing resource utilization and reducing waste.

Looking ahead, future directions for gas flaring management in Nigeria include further research and innovation in technology, policy, and stakeholder engagement. Continued investment in research and development will advance the capabilities of machine learning models and improve the accuracy of flaring predictions. Policy reforms and regulatory frameworks will need to be strengthened to incentivize sustainable practices and ensure compliance across the industry. Moreover, ongoing collaboration and engagement with key stakeholders will be essential for driving collective action towards achieving sustainability goals and addressing climate change challenges in Nigeria. Through concerted efforts and a commitment to data-driven decision-making, Nigeria can pave the way for a more sustainable and environmentally responsible future.

### Future Research

The future of gas flaring management lies in continued research and development efforts aimed at refining machine learning (ML) models, exploring new data sources, and advancing flaring mitigation technologies. One avenue for future studies involves refining ML models to enhance their predictive accuracy and robustness. This could involve incorporating additional variables, such as atmospheric conditions, wind patterns, and geological features, to improve the precision of flaring predictions. Moreover, advancements in data processing techniques, such as deep learning and ensemble methods, could further enhance the performance of ML models in gas flaring estimation.

Exploring new data sources is another promising area for future research. In addition to satellite imagery and operational data from oil & gas companies, emerging technologies like drones and IoT sensors offer new opportunities for real-time monitoring of flaring activities. Integrating data from these diverse sources and leveraging cloud computing and big data analytics will enable more comprehensive and accurate assessments of flaring volumes and trends. Advancing flaring mitigation technologies is crucial for achieving sustainable energy practices and reducing environmental impacts. Future research could focus on developing innovative technologies for capturing and utilizing flare gas, such as micro-scale liquefaction systems, modular gas-to-liquid (GTL) plants, and distributed power generation solutions. Additionally, research into novel carbon capture and storage (CCS) techniques and renewable energy alternatives can help transition towards cleaner and more sustainable energy systems. Collaboration with research institutions and industry partners will be essential for driving innovation and translating research findings into practical solutions. By fostering interdisciplinary collaboration and knowledge sharing, researchers can accelerate the development and adoption of cutting-edge technologies and best practices in gas flaring management. Public-private partnerships, research consortia, and collaborative initiatives can facilitate resource pooling, joint research projects, and technology transfer, ultimately driving positive change in the industry.

### **C. Call to Action**

Gas flaring remains a pressing environmental and socio-economic issue with far-reaching implications for sustainable development. As signatories to the 2030 Agenda for Sustainable Development, nations are committed to achieving various Sustainable Development Goals (SDGs), including SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities and Communities), SDG 14 (Life Below Water), and SDG 3 (Good Health and Well-being). However, the continued practice of gas flaring undermines these goals by contributing to air pollution, climate change, and adverse health effects on communities.

To address this challenge, urgent action is needed at both the national and global levels. Governments, industry stakeholders, research institutions, and civil society must work together to develop and implement effective policies, technologies, and practices for reducing gas flaring. This includes setting ambitious targets for flaring reduction, strengthening regulatory frameworks, incentivizing investment in flare gas recovery technologies, and promoting sustainable energy alternatives. Dynamic production optimization further enhances sustainability efforts by utilizing prescriptive models to optimize production processes. These models consider factors like production rates, equipment performance, and environmental regulations to generate optimal operating strategies. By simulating different scenarios and evaluating potential outcomes, prescriptive models identify the most effective course of action to minimize gas flaring while maximizing efficiency. Adaptive control techniques enable continuous adjustment of operating parameters in response to changing conditions, ensuring that production processes remain optimized in real-time.

In addition to traditional approaches, leveraging machine learning (ML) and artificial intelligence (AI) can revolutionize gas flaring management. These technologies enable real-time monitoring and predictive analytics, allowing for proactive intervention and optimization of production processes. By integrating ML and AI into regulatory frameworks and industry practices, we can enhance efficiency, minimize environmental impact, and redirect resources towards more sustainable development initiatives.

Furthermore, there are promising technologies emerging in the field of gas utilization that offer alternatives to flaring. Small-scale LNG plants, gas-to-liquids (GTL) technology, microgrid applications, enhanced oil recovery (EOR), and plasma conversion are just a few examples. These technologies provide avenues for redirecting gas from flaring towards more productive uses, such as power generation, fuel production, and even fertilizer synthesis. As these technologies continue to mature and become more cost-effective, we can expect to see a significant reduction in the amount of gas that is flared globally. By embracing innovative solutions, investing in research and development, and committing to sustainable practices, we can collectively work towards a future where gas flaring is minimized, energy is produced responsibly, and communities thrive in a clean and healthy environment.

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## **Appendix**

Please find the machine learning model IPython Notebook file at:

[Stse Machine learning code.ipynb - Colaboratory \(google.com\)](#)